HDOC Day-71 – C Programming

1. Aim

To write a C program to dynamically allocate two matrices and perform addition, subtraction, and multiplication using pointer notation and separate functions, and properly release the allocated memory.

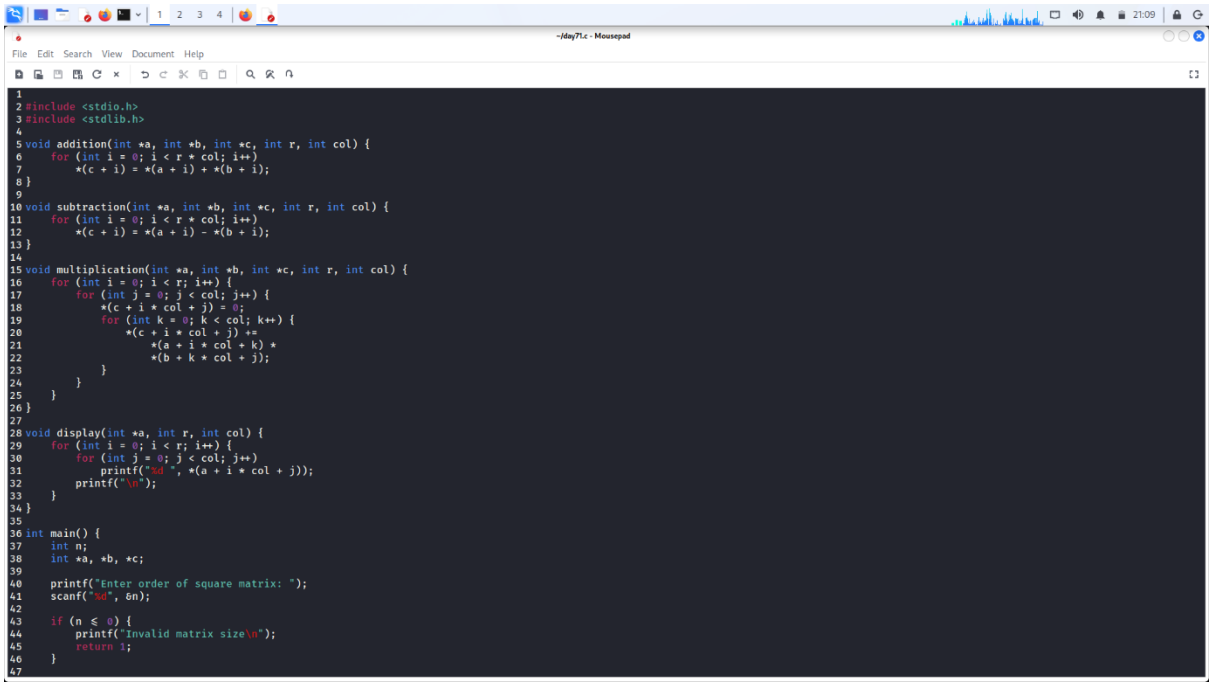
2. Problem Statement

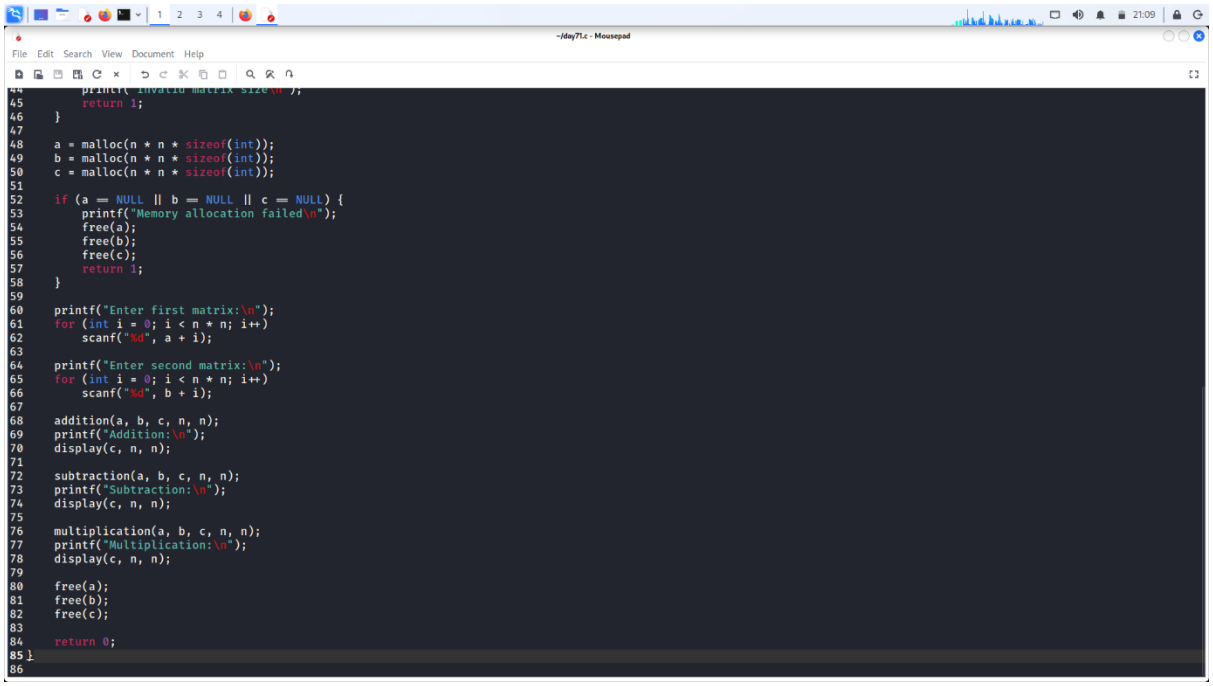
Write a C program to dynamically allocate two matrices and perform addition, subtraction, and multiplication using pointer notation. Implement each operation as a separate function and release all allocated memory properly.

3. Algorithm

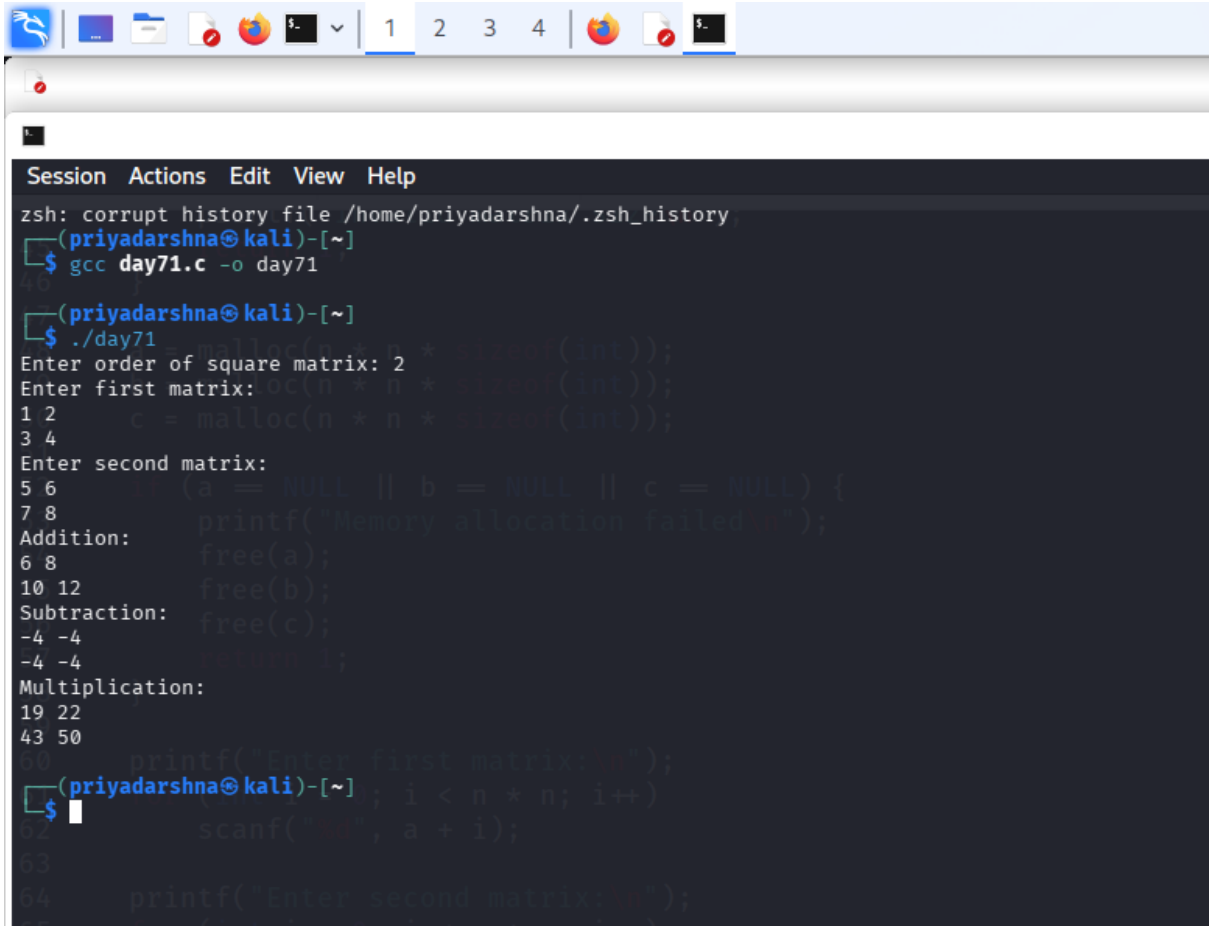
1. Start the program.
2. Read the number of rows and columns.
3. Dynamically allocate memory for two matrices using malloc().
4. Read the elements of both matrices using pointers.
5. Create separate functions for matrix addition, subtraction, and multiplication.
6. Display the results of all three operations.
7. Release the allocated memory using free().
8. Stop the program.

>C PROGRAM:




~/day71.c - Mousepad
File Edit Search View Document Help
printf("Invalid matrix size\n");
return 1;
}
a = malloc(n * n * sizeof(int));
b = malloc(n * n * sizeof(int));
c = malloc(n * n * sizeof(int));
if (a == NULL || b == NULL || c == NULL) {
printf("Memory allocation failed\n");
free(a);
free(b);
free(c);
return 1;
}
printf("Enter first matrix:\n");
for (int i = 0; i < n * n; i++)
scanf("%d", a + i);
printf("Enter second matrix:\n");
for (int i = 0; i < n * n; i++)
scanf("%d", b + i);
addition(a, b, c, n, n);
printf("Addition:\n");
display(c, n, n);
subtraction(a, b, c, n, n);
printf("Subtraction:\n");
display(c, n, n);
multiplication(a, b, c, n, n);
printf("Multiplication:\n");
display(c, n, n);
free(a);
free(b);
free(c);
return 0;
}

>OUTPUT:


Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna㉿kali)-[~]
$ gcc day71.c -o day71
(priyadarshna㉿kali)-[~]
$ ./day71
Enter order of square matrix: 2
Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8
Addition:
6 8
10 12
Subtraction:
-4 -4
-4 -4
Multiplication:
19 22
43 50
(priyadarshna㉿kali)-[~]
$